

ANIKET GHOSH — WEBSITE CONTENT (UPDATED)

Based on latest resume - January 2026

SECTION 0: META & NAVIGATION

Page Title

Aniket Ghosh | AI/ML Engineer & Researcher

Navigation Links

01. About

02. Experience ← UPDATED (you have real experience now)

03. Research

04. Projects

05. Skills

06. Contact

Social Links

GitHub:

https://github.com/Itsme-aniketghosh

LinkedIn:

https://linkedin.com/in/aniketghosh-

Email:

ghosh.anik@northeastern.edu

CodeChef:

https://www.codechef.com/users/a_ghosh (4★)

HackerRank:

https://www.hackerrank.com/profile/aniket_ghosh

Location:

Boston, MA, USA

SECTION 1: HERO

Pre-text

Hi, my name is

Name

Aniket Ghosh.

Typewriter Titles

```
const titles = [  
  "AI/ML Engineer",  
  "NLP Teaching Assistant",  
  "Multi-Agent Systems Builder",  
  "MS AI @ Northeastern",  
  "Healthcare AI Researcher",  
  "Computer Vision Engineer",  
  "MLOps Engineer",  
  "Graduate Researcher"  
];
```

Hero Description

RECOMMENDED:

First-year grad student learning the ropes at Northeastern's Khoury College. TAing NLP, sitting in on a PhD interpretability seminar, and spending too much time on side projects involving multi-agent systems. Always happy to chat about what I'm working on.

Hero Buttons

Primary: "View My Work" → #projects

Secondary: "Get In Touch" → mailto:ghosh.anik@northeastern.edu

SECTION 2: ABOUT ME (01.)

Paragraph 1 — Origin & Values

My path into AI started with contrasts. Growing up, I traveled with my father – an SAP consultant at IBM – through cities like Chicago, Bonn, and Zurich. At the same time, I spent seven years teaching mathematics and computer science to underprivileged students through The Hope Foundation. That gap between what technology could do and who it actually reached became the question I've been trying to answer ever since.

Paragraph 2 — Academic Journey

I graduated with a B.Tech in Computer Science & Business Systems from the Institute of Engineering & Management, Kolkata – ranking 2nd in a department of 180+ students with a 9.3/10 GPA and receiving the Director's Award. In Fall 2025, I joined Northeastern University's Khoury College for my MS in Artificial Intelligence with a 15% merit scholarship, focusing on the Machine Learning concentration.

Paragraph 3 — What I Do Now

Currently, I'm a Teaching Assistant for the graduate NLP course under Professor Amir Tahmasebi, where I teach PyTorch and ML/DL labs. My coursework spans Foundations of AI, Algorithms, Interpretable AI, and Applied Programming for AI. Outside class, I'm deep into multi-agent systems – building autonomous trading simulations with 4 AI agents and

44 integrated tools, and benchmarking link prediction methods on biomedical knowledge graphs.

Technologies List (Updated from Resume)

- Programming Languages:
- Python, C/C++, Java, SQL, HTML/CSS
- LLM & Generative AI:
- RAG, LoRA/QLoRA, Fine-tuning, Prompt Engineering
 - OpenAI API, Anthropic API, Hugging Face, OpenRouter
 - Function Calling, Multi-modal AI, Embeddings
- AI Agents & Orchestration:
- OpenAI Agents SDK, CrewAI, LangChain, LangGraph, AutoGen
 - Model Context Protocol (MCP), Agentic Workflows, Multi-Agent Systems
- ML Frameworks & Libraries:
- PyTorch, TensorFlow, Keras, scikit-learn, Transformers
 - YOLOv8, CNNs, U-Net, Mask R-CNN, Bert
- Vector Databases & Knowledge:
- FAISS, ChromaDB, NetworkX, Knowledge Graphs
- MLOps & Cloud:
- AWS (Bedrock, SageMaker, Lambda, S3, CloudFront, API Gateway)
 - Azure, GCP, Vercel, Docker, Kubernetes
 - Terraform, GitHub Actions, CI/CD, MLflow
- Web & APIs:
- FastAPI
- Specializations:
- Computer Vision, NLP, Medical Image Analysis
 - Deep Learning, Explainable AI, Model Monitoring, Guardrails

Stats Boxes

9.3 CGPA (B.Tech)	2nd Dept Rank / 180+
TA NLP (Grad)	7 yrs Volunteer Teaching

Alternative stats:

9.3 GPA | 2nd/180 Rank | NLP TA | 7 Years Volunteer Teaching

SECTION 3: EXPERIENCE (02.) — NEW SECTION

Experience Card 1: Teaching Assistant ★ CURRENT

ROLE:

COMPANY:

DURATION:

LOCATION:

Teaching Assistant – NLP (Graduate Course)

Northeastern University, Khoury College

Fall 2025 – Present

Boston, MA

DESCRIPTION:

Teaching PyTorch and ML/DL labs for Professor Amir Tahmasebi's graduate NLP course. Helping students understand deep learning fundamentals, transformer architectures, and practical implementation skills. (also grading)

HIGHLIGHTS:

• Teaching PyTorch fundamentals and neural network implementation

• Running hands-on ML/DL lab sessions

• Supporting graduate students with assignments and projects

TAGS:

PyTorch | NLP | Deep Learning | Teaching

Experience Card 2: Research Mentor

ROLE:

COMPANY:

DURATION:

LOCATION:

Researcher & Research Mentor

Jadavpur University

Mar 2024 – May 2025

Kolkata, India

DESCRIPTION:

Developed custom ML pipelines for cell segmentation in histopathological images under Dr. Kaushiki Roy and Dr. Debotosh Bhattacharjee. Implemented

preprocessing techniques using color quantization and DBSCAN clustering for densely packed cell detection.

HIGHLIGHTS:

- Built ML pipelines combining classical CV and deep learning approaches
- Implemented color quantization and DBSCAN clustering for cell detection
- Guided junior researchers in deep learning for medical imaging
- Conducted literature review sessions for the research team

TAGS: Medical Imaging | PyTorch | Computer Vision | Mentorship

Experience Card 3: Research Intern

ROLE: Research Intern
COMPANY: North-Eastern Hill University (NEHU)
DURATION: Feb 2023 – Jul 2023
LOCATION: Remote

DESCRIPTION:

Contributed to developing an Indian Sign Language dataset under Dr. Arnab Kumar Maji. Defined quality criteria and annotation standards by analyzing existing datasets including Microsoft ASL, MNIST, and Static ISL to establish best practices.

HIGHLIGHTS:

- Defined quality criteria and annotation standards for ISL dataset
- Analyzed Microsoft ASL, MNIST, Static ISL datasets
- Established best practices for data collection

TAGS: Computer Vision | Dataset Development | Sign Language | Research

Experience Card 4: Volunteer Educator

ROLE: Volunteer Educator & Mentor
COMPANY: The Hope Foundation
DURATION: 2006 – May 2025 (7 years active teaching)
LOCATION: Kolkata, India

DESCRIPTION:

Taught mathematics and computer science to 10 underprivileged students over seven years. Developed custom learning materials and structured lesson plans that broke complex topics into manageable concepts. Mentored students beyond academics.

HIGHLIGHTS:

- 7 years of active teaching to underprivileged students
- Developed custom learning materials for math and CS
- Created structured lesson plans for complex topics
- Provided mentorship beyond academic subjects

TAGS: Teaching | Mentorship | Social Impact | Education

Experience Card 5: Industrial Trainee

ROLE: Industrial Trainee

COMPANY: Novotel Kolkata Hotel & Residences

DURATION: Dec 2022 – Jan 2023

LOCATION: Kolkata, India

DESCRIPTION:

Facilitated IT system migration from local servers to Oracle cloud infrastructure. Supported IT operations for 1000+ room property.

TAGS: Cloud Migration | Oracle | IT Infrastructure

SECTION 4: RESEARCH & PUBLICATIONS (03.)

Research Card 1: Cell Segmentation (NO PAPER — Corrected)

STATUS: Research Project (Jadavpur University)

ICON: fa-microscope

TITLE: ML Pipeline for Histopathological Cell Segmentation

DESCRIPTION:

Designed custom ML pipeline for histopathological cell segmentation combining classical computer vision and deep learning approaches under Dr. Kaushiki Roy and Dr. Debotosh Bhattacharjee at Jadavpur University.

Implemented preprocessing techniques using color quantization and DBSCAN clustering for detecting densely packed cellular structures. Also mentored junior researchers in deep learning methodologies for medical imaging applications.

TAGS: PyTorch | Medical Imaging | DBSCAN | Computer Vision | Deep Learning

NOTE: Research project – not a publication

Research Card 2: Traffic Sign Detection

STATUS: Published – AISC 2024

ICON: fa-traffic-light

VENUE: Springer "Information and Communication Technologies" series

TITLE: Intelligent Traffic Sign Detection System

DESCRIPTION:
Integrated YOLOv8 with a custom CNN filtering layer to minimize false positives in cluttered Indian driving environments. The joint detection-filtering pipeline significantly improved accuracy over baseline YOLO in challenging real-world conditions.

Published in Springer's "Information and Communication Technologies" series at AISC 2024.

TAGS: YOLOv8 | CNN | Computer Vision | Transfer Learning | Python

LINKS: <https://aisc.bppimt.ac.in/>

Research Card 3: Network Intrusion Detection

STATUS: Under Development – Patent Pending

ICON: fa-shield-alt

TITLE: Hybrid Network Intrusion Detection System

DESCRIPTION:
Designing a network-wide behavioral analysis approach that detects coordinated threats invisible to traditional node-by-node inspection. Developing a hybrid deep learning approach that integrates multiple architectures for enhanced threat detection.

TAGS: Cybersecurity | Deep Learning | Behavioral Analysis | Network Security

LINKS: [Details restricted – patent pending]

SECTION 5: PROJECTS (04.) — GITHUB PROJECTS

Project Card 1: Biomedical Link Prediction ★ FEATURED

ICON:

fa-dna or fa-project-diagram

LINKS:

<https://github.com/Itsme-aniketghosh/biomedical-link-prediction>

TITLE:

Biomedical Knowledge Graph Link Prediction

CONTEXT:

Course project for CS 5100 (Foundations of AI) at Northeastern under Professor Amir Tahmasebi

DESCRIPTION:

Comparative study of link prediction methods on the BioRED biomedical knowledge graph. Benchmarked 7 models and discovered that classical ML dramatically outperforms state-of-the-art knowledge graph embeddings on sparse biomedical data:

- Random Forest: 0.94 AUC (WINNER)
- Gradient Boost: 0.93 AUC
- ComplEx: 0.71 AUC
- TransE: 0.61 AUC

Implemented multi-hop reasoning that traces paths through intermediate biological entities, providing human-readable explanations for every high-confidence prediction.

KEY INSIGHT:

"Simple beats complex on sparse graphs. Structural features like preferential attachment and PageRank capture fundamental patterns that embedding models struggle to learn with limited data."

TAGS:

Knowledge Graphs | PyTorch | scikit-learn | NetworkX | BioRED

METRICS:

7 models | 0.94 AUC | Explainable multi-hop reasoning

Project Card 2: Autonomous Traders MCP ★ FEATURED

ICON:

fa-chart-line or fa-robot

LINKS:

<https://github.com/Itsme-aniketghosh/autonomous-traders-mcp>

TITLE:

Autonomous Multi-Agent Trading Simulation

DESCRIPTION:

Sophisticated equity trading simulation featuring 4 autonomous AI traders making real investment decisions, powered by the Model Context Protocol (MCP).

THE TRADERS:

- Warren – Value investing, patient, fundamental analysis
- George – Macro trading, aggressive, contrarian
- Ray – Risk parity, systematic, diversified
- Cathie – Disruptive innovation, bold, high-growth

ARCHITECTURE:

- 6 MCP servers providing 44 integrated tools
- OpenAI Agents SDK for agent orchestration
- Researcher sub-agent for market analysis
- Gradio dashboard for real-time monitoring
- SQLite persistence + Polygon.io market data

Each trader starts with \$10,000 and manages their portfolio according to their distinct investment philosophy.

TAGS: MCP | OpenAI Agents SDK | Multi-Agent | Gradio | Python

METRICS: 4 agents | 6 MCP servers | 44 tools | Real-time dashboard

Project Card 3: CrewAI Multi-Agent Collection

ICON: fa-users-cog

LINKS: <https://github.com/Itsme-aniketghosh/crewai>

TITLE: CrewAI Multi-Agent Systems Collection

DESCRIPTION:

5 complete multi-agent projects demonstrating collaborative AI:

1. ENGINEERING TEAM – 4 agents (Lead → Backend → Frontend → Test) delivering working code with tests and UI
2. STOCK PICKER – Hierarchical manager with long-term memory, short-term memory, and custom push notification tools
3. DEBATE – Multi-LLM debate simulation (GPT-4o-mini vs Claude)
4. FINANCIAL RESEARCHER – Web-powered company analysis
5. CODER – Docker-based safe code execution

TAGS: CrewAI | Multi-Agent | GPT-4o | Claude | AutoGen | LangChain

METRICS: 5 projects | 15+ agents | Multi-LLM orchestration

Project Card 4: LLM Cybersecurity Analyzer

ICON:

fa-shield-virus

LINKS:

<https://github.com/Itsme-aniketghosh/LLM-Cybersecurity-Analyzer>

TITLE:

AI-Powered Code Security Analyzer

DESCRIPTION:

Full-stack application for automated Python vulnerability detection combining OpenAI Agents SDK with Semgrep static analysis.

TECH STACK:

- Frontend: Next.js + TypeScript + Tailwind CSS
- Backend: FastAPI + Python 3.12
- AI: OpenAI Agents SDK + Semgrep MCP server
- Deployment: Azure Container Apps + GCP Cloud Run
- IaC: Terraform with multi-cloud workspaces

Built as part of Ed Donner's "AI in Production" course.

TAGS:

OpenAI Agents | Semgrep | Next.js | FastAPI | Terraform

METRICS:

Full-stack | Multi-cloud | Production-ready

Project Card 5: DeepResearch

ICON:

fa-search

LINKS:

<https://github.com/Itsme-aniketghosh/LLM-Deep-Research> [Live Demo]

TITLE:

AI Research Assistant

LIVE DEMO:

<https://ag-hosh-deeprerearch.hf.space/>

DESCRIPTION:

Autonomous research system with multi-agent pipeline:

- Planner Agent – Generates 3-7 search queries
- Parallel Research Agents – Extract hooks, facts, insights
- Writer Agent – Synthesizes 3 journalistic paragraphs
- Research Manager – Orchestrates asynchronously

TAGS:

Llama 3.2 | Multi-Agent | asyncio | Gradio | HuggingFace

METRICS:

Live on HuggingFace | Real-time streaming

Project Card 6: Digital Twin RAG

ICON:fa-user-circle

LINKS:<https://github.com/Itsme-aniketghosh/LLM-my-digital-twin> [Live Demo]

TITLE:RAG-based Digital Twin

LIVE DEMO:<https://ag-hosh-my-digital-twin.hf.space/>

DESCRIPTION:
Production RAG system serving grounded, first-person responses over personal documents.

- Embeddings: all-MiniLM-L6-v2
- Vector Store: FAISS
- LLM: Llama 3.1 8B Instruct
- Chunking: 600 words / 200 overlap

TAGS:RAG | FAISS | Llama 3.1 | ChromaDB | Gradio

METRICS:Sub-ms retrieval | 2-5s E2E latency

SECTION 6: SKILLS (05.) — UPDATED FROM RESUME

Category 1: Programming Languages

Python	95%
C / C++	85%
Java	80%
JavaScript	75%
SQL	85%
HTML/CSS	70%

Category 2: LLM & Genertic AI

RAG & Vector DBs (ChromaDB)	90%
OpenAI/Anthropic APIs	90%
Hugging Face & Transformers	85%

LoRA/QLoRA Fine-tuning	75%
OpenRouter	80%

Category 3: AI Agents & Orchestration

OpenAI Agents SDK	90%
CrewAI	90%
Model Context Protocol (MCP)	85%
LangChain / LangGraph	85%
AutoGen	80%
Function Calling	85%

Category 4: ML Frameworks

PyTorch	95%
TensorFlow / Keras	90%
scikit-learn	90%
YOLOv8 & CNNs	90%
Transformers	85%
U-Net	80%

Category 5: MLOps & Cloud

AWS (Bedrock, SageMaker, S3)	80%
Azure	75%
GCP	75%
Docker	85%
Terraform	75%
GitHub Actions	80%
MLflow	75%

Category 6: Specializations

Computer Vision	90%
Deep Learning	90%
Medical Image Analysis	85%
NLP	85%
Explainable AI	80%



SECTION 7: CERTIFICATIONS (Optional Section)

```
\textbf{Completed Online Courses with Certificates}
\begin{itemize}[noitemsep, topsep=0pt, partopsep=0pt, parsep=0pt]
  \item \textbf{Deep Learning \& AI:}
  \href{https://www.udemy.com/certificate/UC-17f58812-6746-49db-9f1b-55288d0f4cc8/}{Deep Learning A-Z 2024}, \href{https://www.udemy.com/certificate/UC-82fb2982-1ef2-4612-8dce-85c6247f18d7/}{Machine Learning: NLP in Python V2}
  \item \textbf{ML Frameworks \& Data Science:}
  \href{https://www.udemy.com/certificate/UC-057b3f62-2d2f-4ac8-88b7-4bb3ac32099e/}{Python for ML \& Data Science Masterclass}
  \item \textbf{LLM \& GenAI Engineering:}
  \href{https://www.udemy.com/certificate/UC-6da190e8-0ec9-4582-b1f9-a3d6e7ef9fd0/}{AI Engineer Core Track}, \href{https://www.udemy.com/certificate/UC-a96b7f21-f370-4bf6-b298-68afbf573acb/}{AI Engineer Agentic Track},
  \href{[https://www.udemy.com/course/generative-and-agentic-ai-in-production/]}{MLOps: Deploy to AWS/Azure/GCP}
  \item \textbf{Programming \& Algorithms:}
  \href{https://www.udemy.com/certificate/UC-d732f376-d464-4c4e-85cc-fcf11e7216f8/}{Data Structures \& Algorithms C++}, \href{https://www.udemy.com/certificate/UC-fedcd3f8-b13a-4412-acaf-4c56a217f4fe/}{C++ Programming}
  \item \textbf{Specialized Skills:} \href{https://www.udemy.com/certificate/UC-7a1d8574-7ee2-4c0a-9ed7-6b2b6e157270/}{High-Performance Computing with Python 3x},
  \href{https://www.udemy.com/certificate/UC-49132f6d-e26a-4b6c-b43b-9e0d57b234b2/}{Investing in Stocks Complete Course}
\end{itemize}
```

SECTION 8: CONTACT (06.)

Section Label

06. What's Next?

Title

Get In Touch

Description

I'm currently pursuing my MS in AI at Northeastern University (graduating May 2027) while TAing the graduate NLP course. I'm actively seeking Summer 2026 internships in ML Engineering, AI Research, or Applied AI roles.

Whether you have an opportunity, a research collaboration idea, or just want to chat about multi-agent systems – my inbox is always open.

CTA Button

[Say Hello] → <mailto:ghosh.anik@northeastern.edu>

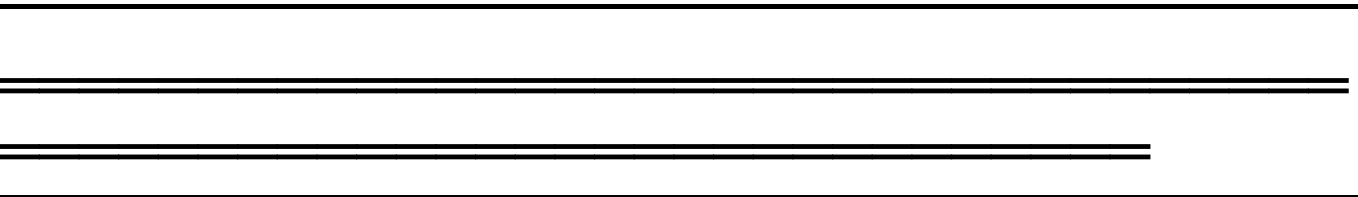
Contact Details

 ghosh.anik@northeastern.edu

 [linkedin.com/in/aniketghosh-](https://www.linkedin.com/in/aniketghosh-)

 github.com/Itsme-aniketghosh

 Boston, MA, USA



SECTION 9: FOOTER

Designed & Built by Aniket Ghosh • 2026

[\[GitHub\]](#) [\[LinkedIn\]](#) [\[CodeChef\]](#) [\[HackerRank\]](#)